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كلية علوم الحاسب والمعلومات
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IT362 Course Project

Semester 2, 1447 H

SENTIMET ANALYSIS OF REVIEWS FOR MENTAL HEALTH APP

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1. INTRODUCTION

Mental health applications are widely used to support emotional well-being, yet understanding users' true experiences remains challenging. While app ratings provide a general measure of satisfaction, they fail to capture the emotions and detailed opinions expressed in written reviews.

This project addresses this limitation by applying sentiment analysis to user reviews of mental health applications in order to identify overall user sentiment.

The main research question is:

How can sentiment analysis of user reviews help in understanding users' perceptions of mental health applications?

2. LITERATURE REVIEW

This section reviews relevant research on sentiment analysis applied to mental health and health-related mobile applications. Five key studies were selected based on their relevance to app review sentiment analysis, methodological rigor, and contribution to the field.

Study 1: Manual Annotation Based Sentiment Analysis of Health App Reviews

Reference:

Varghese, L., Pai, R. R., Savitha, G., Girisha, S., & Chetty, N. (2025). Manual annotation based sentiment analysis of user feedback in health and wellness app reviews. Scientific Reports, 15, 44766. https://doi.org/10.1038/s41598-025-28799-5

Problem Addressed:

- Absence of publicly available, manually annotated datasets for health app sentiment analysis
- Severe class imbalance in real-world app review data (extreme positive/negative sentiments dominate)
- Lack of systematic evaluation of multiple ML and DL models across rebalanced datasets

Dataset Used:

- Source: Google Play Store reviews of HealthifyMe (Health and Wellness app)
- Total size: 20,651 manually annotated reviews
- Collection period: April 30, 2021 to July 5, 2024
- Language: English only
- Region: Indian users

Methods/Models Applied:

• Manual annotation of reviews into five sentiment classes
• Multi-class sentiment classification framework
• Comparative evaluation of multiple advanced machine learning models
• Emphasis on human-labeled data to enhance model reliability

Key results and findings:

• Manual annotation significantly improved sentiment classification accuracy
• Advanced models outperformed baseline approaches in multi-class sentiment tasks
• The proposed approach effectively captures nuanced emotional feedback
• Findings demonstrate applicability for improving user experience in health-related app

Study 2: User Reviews of Depression App Features

Reference:

Meyer, J., & Okuboyejo, S. (2021). User Reviews of Depression App Features: Sentiment Analysis. JMIR Formative Research, 5(12), e17062. https://doi.org/10.2196/17062

Problem Addressed:

- Mental health conditions remain undertreated with barriers to accessing care
- Little known about emotional reactions of depression app users to different app features
- Emotions are key component of mental health but understudied in mHealth apps
- Need to understand which app features generate positive vs. negative emotional responses
- User perceptions decisive in mHealth app adoption but not systematically analyzed

Dataset Used:

- Source: Google Play Store and Apple App Store
- Total reviews: 3,261 user reviews
- Number of apps: 61 depression-related apps
- Collection date: March 2018
- Language: English

Methods/Models Applied:

- Linguistic Inquiry Word Count (LIWC) 2015
- Four Summary Dimensions Analyzed (0-100 scale)
- Emotion Subcategories Analyzed
- Word Cloud Generation
- Statistical Analysis

Key results and findings:

The findings showed that user sentiment significantly differed across app feature categories. Psychoeducational and therapeutic features were generally associated with more positive emotional expressions, while entertainment features showed more mixed sentiments.
The study demonstrated that feature-level sentiment analysis is an effective approach for understanding user experiences and improving depression app development

Study 3: Mental Health Screening Apps User Perspectives

Reference:

Huckvale, K., Nicholas, J., Torous, J., & Larsen, M. E. (2022). mHealth Solutions for Mental Health Screening and Diagnosis: A Review of App User Perspectives Using Sentiment and Thematic Analysis. Frontiers in Psychiatry, 13, 857304.
https://doi.org/10.3389/fpsyt.2022.857304

Problem Addressed:

- Understanding user perceptions of mental health screening/diagnostic apps
- Identifying common app features and their reception by users
- Determining which features associated with app discontinuation
- Difficulties in app discovery (finding relevant apps in stores)
- Need to understand barriers to sustained use of mental health apps

Dataset Used:

- Source: Apple App Store, Google Play Store, m-health Index and Navigation Database (MIND)
- · Total apps analyzed: 92 apps
- · Total reviews analyzed: 676 user reviews with metadata

Methods/Models Applied:

Dual Analytical Approach:

- Sentiment Analysis:
  - Classified reviews as Positive, Negative, or Neutral
  - Used automated sentiment classification tools
  - Validated through manual review of samples
- Thematic Analysis:
  - Identified 10 major themes in user reviews
  - Coded reviews for presence of each theme
  - Focused on performance and functionality aspects
  - Analyzed relationship between themes and discontinuatio

Key results and findings:

The results indicated that most user reviews expressed positive sentiment; however, a notable proportion of users reported negative experiences related to app functionality and usability. The study also found cases where numerical ratings did not align with the sentiment expressed in textual reviews, highlighting the importance of analyzing written feedback alongside star ratings.

Study 4: Machine Learning for Health App Reviews

Reference:

Oyebode, O., Alqahtani, F., & Orji, R. (2020). Using Machine Learning and Thematic Analysis Methods to Evaluate Mental Health Apps Based on User Reviews. IEEE Access, 8, 111141-111158. https://doi.org/10.1109/ACCESS.2020.3002176

Problem Addressed:

- Need for systematic evaluation of mental health apps based on real user feedback
- Understanding user experience beyond traditional app ratings
- Identifying key factors influencing user satisfaction and app effectiveness
- Extracting actionable insights from large volumes of unstructured review data
- Bridging gap between clinical features and user-perceived value

Dataset Used:

- Source: App Store and Google Play
- Total reviews: 88,125 user reviews
- Number of apps: 104 mental health apps
- Review diversity: Varying lengths, emoticons, product mentions
- Time period: Multiple years of reviews collected
- Focus: Depression, anxiety, stress management apps

Methods/Models Applied:

- Machine Learning Models
- Deep Learning Architectures:
- Feature Extraction Approaches:
- Bag of Words (BoW)
- TF-IDF weighting
- Word embeddings (Word2Vec, GloVe)
- Aspect extraction (identifying specific features mentioned)·

Study 5: Topic Modeling Analysis of Public Trust in AI Mental Health Applications

Reference:
Author(s). (Year). Topic Modeling Analysis of Public Trust in AI Mental Health Applications. JMIR Human Factors.
Available at: https://pubmed.ncbi.nlm.nih.gov/36459412/

Problem Addressed:
- Limited understanding of public trust in AI-powered mental health applications
- Need to explore whether users trust or distrust AI-based psychological tools
- Lack of thematic analysis focusing specifically on trust-related concerns
- Insufficient research applying topic modeling techniques to mental health app reviews
- Need to identify dominant themes shaping user trust perceptions

Dataset Used:
- Source: Google Play Store
- Total reviews analyzed: 3,931 user reviews
- Number of apps: 8 popular AI-based mental health applications
- Time period: January 2020 – April 2022
- Data type: Unstructured textual user-generated reviews

Methods/Models Applied:

- Web scraping using Python crawler to collect user reviews
- Text preprocessing and cleaning
- Topic Modeling using Latent Dirichlet Allocation (LDA)
- Topic number selection using coherence scores
- Visualization using pyLDAvis
- Multidimensional scaling for clustering and interpreting themes

Key Results and Findings:

- Four dominant topics were identified from user reviews
- Topics were grouped into three major trust-related themes
- Results indicated an overall high level of public trust in AI mental health applications
- Negative reviews highlighted areas requiring improvement
- The study is among the first to apply topic modeling to examine public trust in AI-based mental health apps

COMPARATIVE ANALYSIS OF STUDIES :

1. Common Datasets and Features:

Across all five studies, several patterns emerged in data sources, dataset characteristics, and feature types analyzed:

Common Data Sources:

- Google Play Store (used by ALL 5 studies)
- Apple App Store (used by 4 studies)
- Review metadata: ratings, dates, user info
- Unstructured text as primary data
- Public, user-generated content

Common Features Analyzed:

- Review text content (universal)
- Star ratings (1-5 scale)
- Review timestamp/date
- Review length (words/characters)
- User sentiment (positive/negative/neutral)
- Emotional tone and intensity

App Categories Studied:

- Mental health (depression, anxiety) - Studies 2, 3, 4
- General health and wellness - Study 1
- Screening/diagnostic apps - Study 3
- Therapeutic/treatment apps - Study 2

2. Differences in Modeling Approaches

The four studies employed diverse methodological approaches, reflecting the evolution and variety of sentiment analysis techniques:

- Traditional Machine Learning (Studies 1, 3)
- Deep Learning Approaches (Studies 1, 4)
- Lexicon-Based Methods (Study2)

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3. Strengths of Existing Research:

Methodological Strengths:

- Large-scale datasets (up to 88,000+ reviews)
- Multiple evaluation metrics for robust assessment
- Diverse modeling approaches tested and compared
- Real-world, authentic user feedback analyzed
- Reproducible methodologies with clear documentation
- Cross-validation and rigorous evaluation protocols

4. Limitations of Existing Research:

Data-Related Limitations:

- Limited demographic information about reviewers (age, gender, location)
- No validation of review authenticity (fake reviews possible)
- Platform bias (mostly Android users from Google Play Store)
- Language limitations (English-only)

Scope and Coverage Limitations:

- Limited focus on comprehensive therapy platforms (mostly screening apps)
- Few studies on Arabic-speaking region mental health apps
- Insufficient analysis of Aribc reviews from Middle Eastern markets
- Lack of longitudinal studies tracking sentiment evolution

Research Gaps and Missing Elements

Despite valuable contributions, several critical gaps remain in the literature:
Geographic and Cultural Gaps:

- Minimal research on comprehensive therapy platforms in Middle East
- Limited analysis of apps serving Arabic-speaking populations with English interfaces
- Insufficient understanding of cultural factors affecting mHealth adoption in Saudi Arabia
- Missing comparative studies: Western vs. Middle Eastern user sentiment patterns
- Lack of research on bilingual users (Arabic-English) sentiment patterns

HOW THIS PROJECT ADDRESSES GAPS

This project strategically addresses several critical research gaps identified in the literature review:

- Focuses specifically on Labayh - leading therapy platform in Saudi Arabia
- Analyzes reviews from Arabic-speaking market (underrepresented perspective)
- Provides insights into regional mHealth adoption patterns
- Bridges gap between Western research and Middle Eastern context

HOW THIS PROJECT ADDRESSES GAPS

App Type and Feature Focus:

- Analyzes comprehensive therapy platform (not just screening app)
- Studies app connecting users with licensed therapists (understudied type)
- Examines session booking, payment, and professional service quality aspects
- Investigates privacy concerns specific to therapy platforms
- Focused analysis allows deeper feature-specific insights

3. DATA SOURCES

Primary Source: Google Play Store

Description:

Google Play Store was used to collect publicly available user reviews related to mental health applications. The data represents authentic user feedback from Arabic-speaking users and includes both local and international apps.

Access Method:

Tool: google-play-scraper Python library
Collection: Automated web scraping via API
URL Pattern: https://play.google.com/store/apps/details?id=[APP_ID]
Authentication: No authentication required (public data)

Data Details :

Total Dataset Size: 1,032 reviews

- **Time Range: All available historical comments**
- **Collection Type: One-time data collection (Phase 1)**

Features and Data Types :

Feature Name	Data Type	Description
content	Text (Unstructured)	Arabic comment text - PRIMARY
score	Integer (1-5)	User rating
at	Datetime	Comment timestamp
reviewId	String	Unique identifier
userName	String	User display name
thumbsUpCount	Integer	Helpful votes
replyContent	Text	Developer response
repliedAt	Datetime	Response timestamp
language	String	Detected language
sentiment	Categorical	To be computed (positive/negative/neutral)

3. DATA SOURCES

Target Applications

Target Applications & Collected Reviews

Labayh (لبيه)

• Package ID: com.labayh.app
• Reviews Collected: 537

Estanara (استنارة)

• Package ID: com.app.estenara
• Reviews Collected: 60

Calm

• Package ID: com.calm.android
• Reviews Collected: 425

BetterHelp

• Package ID: com.betterhelp
• Reviews Collected: 10

Potential Biases and Sources

1. REPRESENTATION BIAS

Issue: Not all user groups equally represented
•Active reviewers vs. silent majority
•Smartphone owners only (excludes non-digital users)
•Android users only (no iOS data)
•Users willing to publicly comment (privacy-conscious excluded)
•Tech-literate population bias
Mitigation: Acknowledge limitation; findings represent 'digitally active users' segment

2. MEASUREMENT BIAS

Issue: How data is collected affects results
•Google Play algorithms may filter or prioritize certain comments
•Only publicly posted comments (no private feedback)
•Star ratings may not match text sentiment
•Timing of comment (after good/bad session) affects tone
•Language detection errors (Arabic dialects, mixed text)
Mitigation: Use language detection with validation; acknowledge rating-sentiment mismatches

3. HISTORICAL BIAS

Issue: Perpetuating existing inequalities

- Mental health stigma in Arab culture may suppress negative experiences
- Socioeconomic bias (app users can afford therapy)
- Regional bias .

Mitigation: Contextualize findings

4.LANGUAGE AND CULTURAL BIAS

•Arabic dialects vary (Gulf, Levantine, Egyptian, etc.)
•Formal vs. informal Arabic (Fusha vs. Ammiya)
•Cultural expressions of dissatisfaction (indirect vs. direct)
Mitigation: Use dialect-aware Arabic NLP; manual validation of samples

4. OBJECTIVES

These objectives define the analytical goals and insights the dataset will enable:

OBJECTIVE 1: SENTIMENT CLASSIFICATION

- Classify Arabic comments into positive, negative, and neutral categories
- Analyze sentiment distribution and patterns
- Compare sentiment across different rating levels

OBJECTIVE 2: THEME AND TOPIC EXTRACTION

- Identify common themes in user feedback
- Extract frequently mentioned topics
- Categorize complaints and praise

OBJECTIVE 3: FEATURE-SPECIFIC SENTIMENT

- Link sentiment to specific app features
- Analyze therapist quality, booking, pricing, interface

OBJECTIVE 4: ACTIONABLE INSIGHTS

- Provide recommendations for improvement
- Identify priority areas
- Questions enabled:
- What should developers prioritize?
- How can user experience be enhanced?
- What drives user retention vs. churn?

5. METHOD

How we will achieve the objectives :

Data Collection :

- Use google-play-scraper for automated collection
- Collect in batches (200 comments per batch)
- Implement 1-second delay between requests
- Save raw data (CSV + JSON formats)
- Apply language detection (langdetect library)

Preprocessing :

- Arabic text normalization (remove diacritics)
- Tokenization using Arabic-specific tools
- Remove Arabic stop words
- Handle emojis and special characters
- Clean and deduplicate data

Feature Engineering :

- TF-IDF vectorization for Arabic text
- Arabic Word2Vec embeddings
- N-gram features (unigrams, bigrams)
- Sentiment lexicon scores
- Comment length and rating features

Sentiment Analysis :

- Classification
- Machine Learning
- Deep Learning

Evaluation :

- Accuracy, Precision, Recall, F1-Score
- Confusion matrix analysis
- 80/20 train-test split
- 5-fold cross-validation

6.CHALLENGES AND RECOMMENDATIONS

How we will achieve the objectives:

Language Detection :

- **Issue:** Mixed language text; dialect variations
- **Impact:** Incorrect filtering
- **Solution:** Use langdetect with validation; manual review samples
- **Recommendation:** Use Arabic-specific language detection tools

Arabic Text Complexity:

- **Issue:** Diacritics, dialects, informal text
- **Impact:** Reduced NLP accuracy
- **Solution:** Normalization; dialect-aware tools
- **Recommendation:** Use AraBERT or CAMeL Tools for preprocessing

Limited Arabic Resources:

- **Issue:** Fewer tools than English NLP
- **Impact:** Lower baseline accuracy
- **Solution:** Use available resources (ArSenL, AraBERT)
- **Recommendation:** Contribute findings to Arabic NLP community

General Recommendations:

- Document all steps for reproducibility
- Version control for code and data
- Regular backups during collection
- Manual validation of random samples
- Ethical handling of user data
- Transparent reporting of limitations
- Consider privacy implications
- Plan for future Arabic language expansion

Challenges Faced in Data Collection :

- Limited Arabic Reviews
- The number of Arabic-language reviews was significantly smaller compared to English reviews.
- Platform Restrictions
- Data collection relied on a scraping API, which may not provide access to all available reviews due to platform limitations.
- Language Complexity
- Arabic text processing presents challenges due to dialect variations, spelling inconsistencies, and morphological complexity.
- Rating–Text Inconsistency
- Some reviews show inconsistency between star ratings and textual sentiment.

Logbook – Phase 1

Date: 10 Feb 2026
Task:conducted research and wrote a literature review on it.
Task: Identified data source
• Selected Google Play Store as primary source
• Rationale: Public access + authentic user reviews

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Date: 11 Feb 2026

Task: App Selection
• Selected Labayh ,Estanara ,Calm and better Help
• Added Calm and BetterHelp to increase dataset size
• Rationale: Improve dataset diversity

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Date: 13 Feb 2026

Task: Data Extraction
• Used google-play-scraper
• Applied lang='ar' filter
• Exported raw dataset to CSV
• Total collected: 1,032 reviews
Task: documentationfor the rest of the phase

7. EXPLORATORY DATA ANALYSIS (EDA)

Overview of the Dataset

In this phase, we worked with two datasets related to mental health applications.

The primary dataset was collected from Google Play Store and includes 1,031 Arabic reviews from four applications: Labayh, Estinara, Calm, and BetterHelp. The dataset contains rating (1–5), review text, review date, helpful votes, developer replies, and app version. Some missing values were found in reply-related columns, which is normal because not all reviews receive responses.

The secondary dataset (MHARD) contains 18,000 English reviews collected globally from mental health apps. It is much larger and represents users from different countries and backgrounds. This dataset includes ratings, review text, likes, and time information.

Before starting the analysis, both datasets were cleaned. In the Arabic dataset, we removed duplicates, handled missing values, converted dates into datetime format, cleaned and normalized Arabic text, and created a new feature called text_length to measure the length of each review.

2. Primary Data – EDA Results

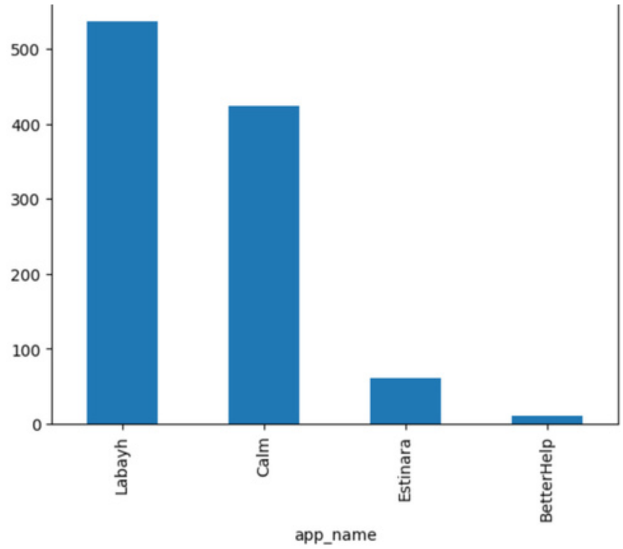
In this section, we explore the primary dataset to understand general patterns in user reviews and ratings for the selected mental health applications. This exploratory analysis helps provide initial insights into user feedback before conducting deeper analysis.

2.1 Reviews per Application

We first looked at how the number of reviews collected per application, we found that the reviews are not evenly distributed across the four applications.

Most of the reviews come from Labayh and Calm, while Estinara has a smaller number of reviews and BetterHelp has the fewest reviews in the dataset.

This means that the dataset is imbalanced, so findings related to apps with fewer reviews should be interpreted more carefully.



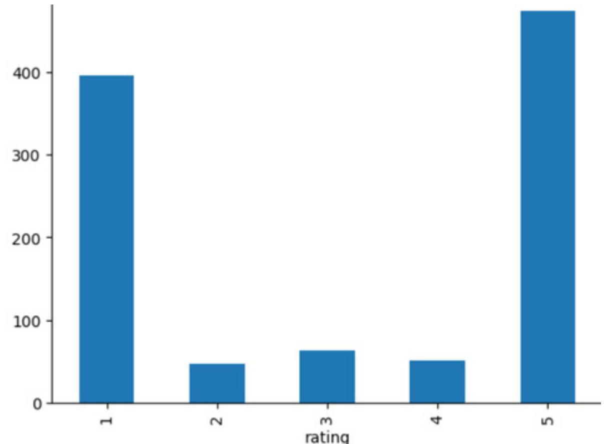
2.2 Rating Distribution

Next, we examined how ratings are distributed across the dataset.

Findings

The ratings show a polarized pattern. Many reviews are 5-star ratings, indicating positive user experiences, but there is also a noticeable number of 1-star ratings.

This suggests that users tend to leave reviews when they have very positive or very negative experiences, rather than neutral ones.



2.3 Average Rating per Application

We then compared the average rating for each app. Estinara had the highest average rating (4.07), followed by Labayh (3.25). Calm and BetterHelp had lower averages within this dataset.

2.4 Review Length Analysis

To further understand user behaviour, we created a new variable review_length, which represents the number of characters in each review. This helps us see whether users tend to write longer reviews when they are satisfied or dissatisfied.

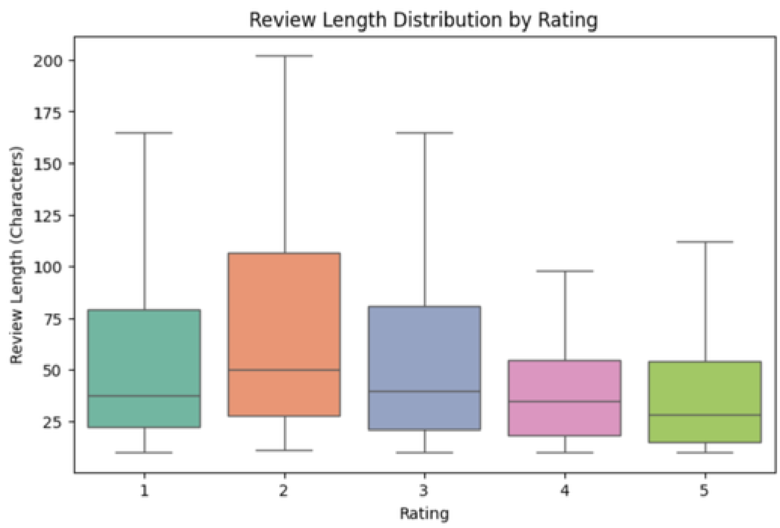
The boxplot shows how review length varies across different rating levels.

Findings

From this analysis, we noticed that:

Lower ratings (1–2 stars) tend to have longer reviews, while higher ratings (4–5 stars) are generally shorter.

This suggests that users often write more detailed explanations when they are dissatisfied.



2.5 Developer Response Rate

We also analyzed how often developers respond to user reviews.

The results show that about 39% of the reviews received a developer response.

Findings

This indicates moderate interaction between developers and users.

However, since many reviews still receive no response, there may be room for improvement in user engagement.

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2.6 Rating Trend Over Time

We also analyzed how the average rating changes over time by calculating the average rating for each year based on the review date.

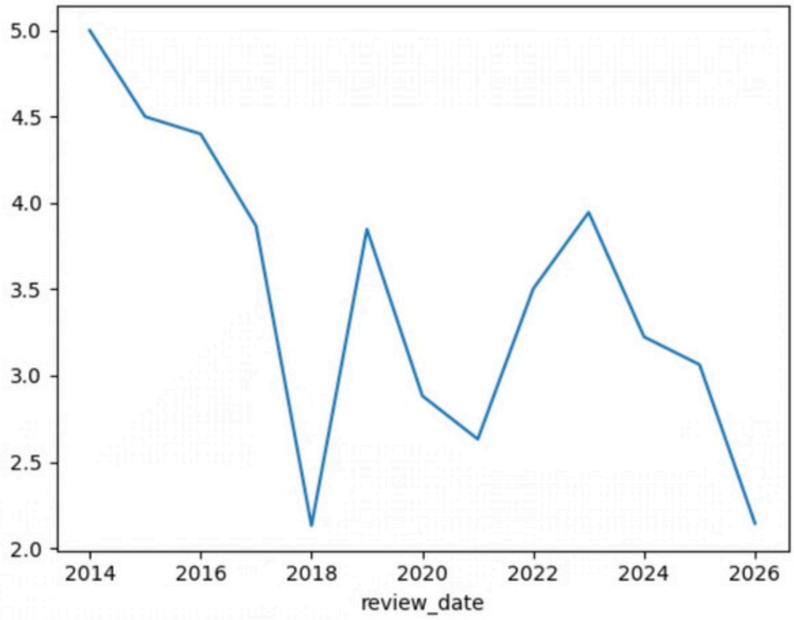
This helps us see whether user satisfaction increased or decreased over the years.

Findings

From the chart, we noticed that:

- The average rating changes from year to year and does not follow a clear trend.
- Some years show higher satisfaction, while others show lower ratings.
- Earlier years have fewer reviews, so their averages may not be very reliable.

Overall, this shows that user satisfaction can vary over time, possibly because of app updates or changes in user experience.



2.7 Relationship Between Rating and Review Length

We measured the correlation between rating and review length.

Findings

The correlation is slightly negative ≈-0.13 , meaning that lower ratings tend to have longer reviews.

However, the relationship is weak, which suggests that review length does not strongly affect the rating.

8.Secondary Data— EDA Results

The secondary dataset represents global users and shows different patterns.

Statistical Summary

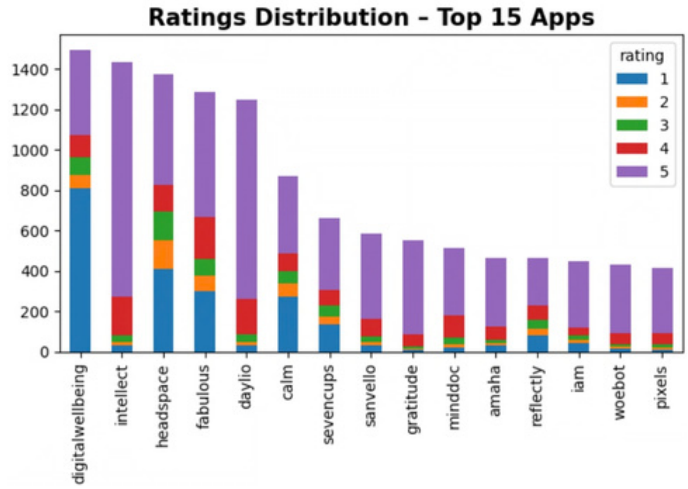
To better understand the dataset, we first examined the statistical summary of the main numerical variables, including rating, likes, and review length.

Findings

The dataset contains 18,000 reviews. The average rating is 3.94 and the median rating is 5, which shows generally positive feedback.
The average review length is about 168 characters, meaning most reviews are short.
For likes, the median number of likes is 0, but a few reviews receive a very high number of likes, with the maximum reaching 2021 likes.

Rating Distribution

Next, we examined how ratings are distributed across the dataset.



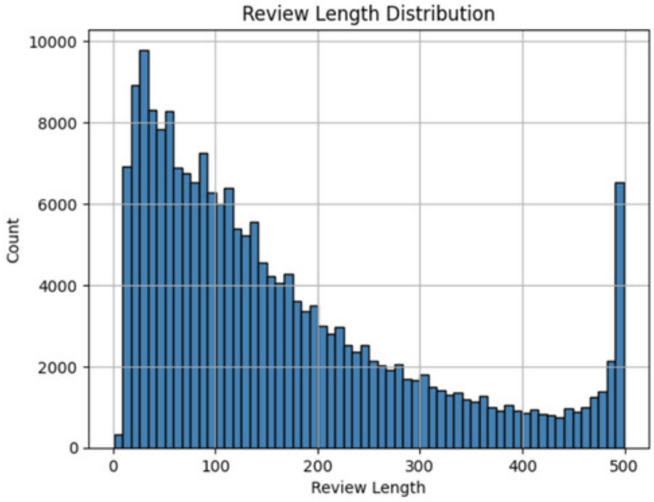
Findings

The rating distribution shows that most reviews are concentrated at 5 stars, indicating a strong positive skew in the dataset.

Ratings of 4 stars also appear frequently, while 2-star and 3-star ratings occur less often.
However, 1-star reviews are still present, suggesting that some users report negative experiences.
Overall, the distribution reflects generally positive user feedback, with a strong concentration of high ratings.

Review Length Distribution

We also analyzed the distribution of review lengths to understand how detailed user feedback tends to be.



Findings

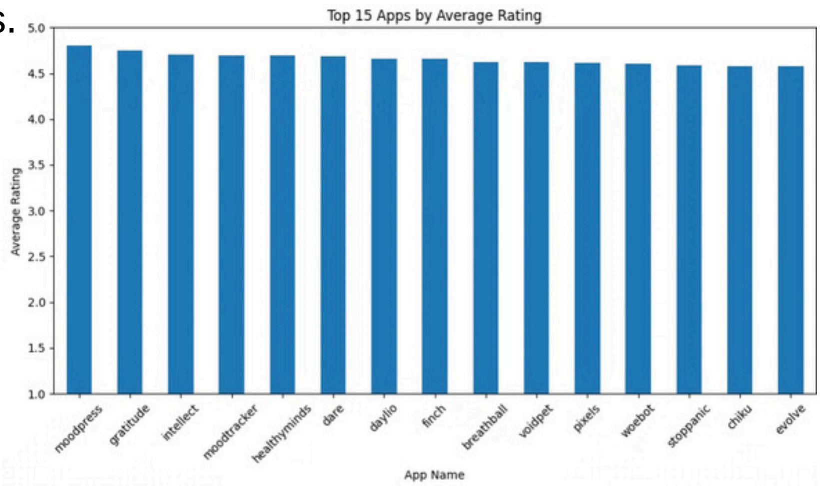
The distribution is right-skewed, which means most reviews are short.

Most reviews are under 200 characters, while longer reviews appear less often.

Overall, this shows that most users write short reviews, while fewer users provide longer and more detailed feedback.

Review per Application

To understand how reviews are distributed across apps, we counted the number of reviews for each application and visualized the top 15 apps with the highest number of reviews.

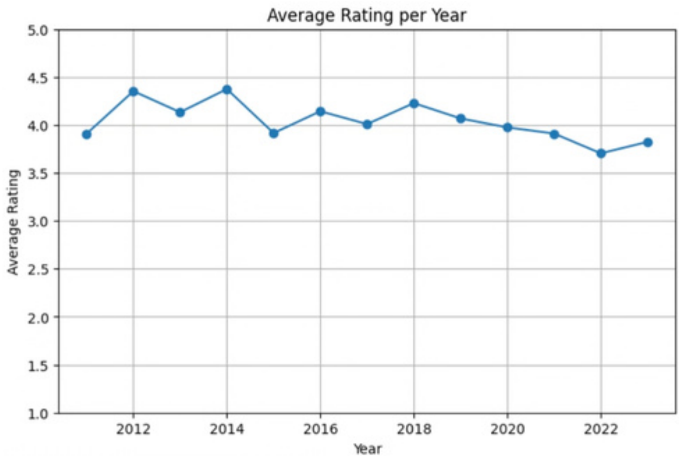


Findings

The chart shows that the top 15 apps have very high average ratings, mostly between 4.6 and 4.8, indicating strong user satisfaction. The ratings are very close across apps, suggesting that these platforms provide a similar level of quality and user experience.

Average Rating per Year

To see how user satisfaction changes over time, we calculated the average rating for each year based on the review dates.



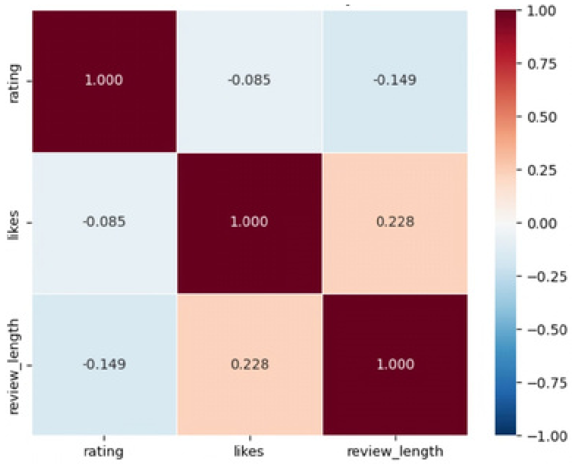
Findings

The ratings change slightly from year to year without a clear increasing or decreasing trend.
Most ratings stay between 4.0 and 4.4, which shows generally positive feedback from users.

Overall, the ratings appear relatively stable over time.

Correlation Between Rating, Likes, and Review Length

We examined the relationship between rating, number of likes, and review length.



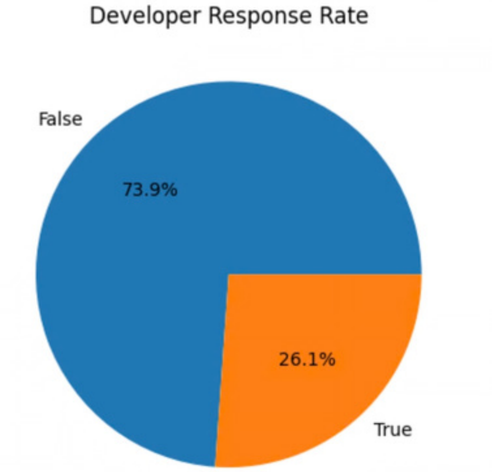
Findings

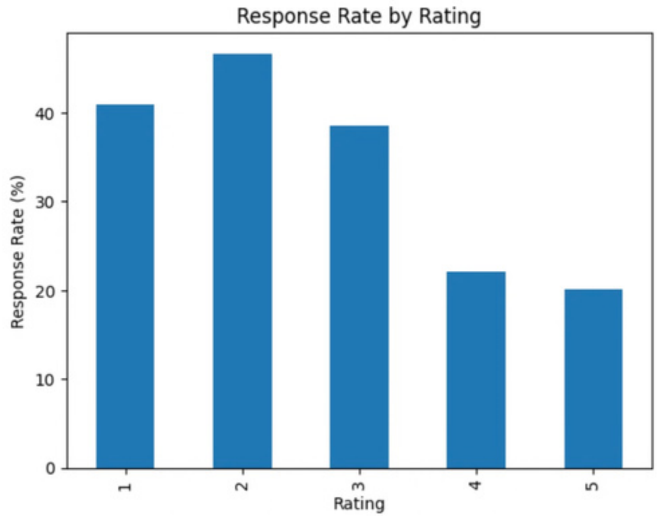
The correlations between these variables are generally weak.
There is a slight negative relationship between rating and review length, meaning longer reviews do not necessarily have higher ratings.

There is also a small positive relationship between review length and likes, suggesting that longer reviews may receive slightly more likes

Developer Response Rate

We also analyzed how often developers respond to user reviews.





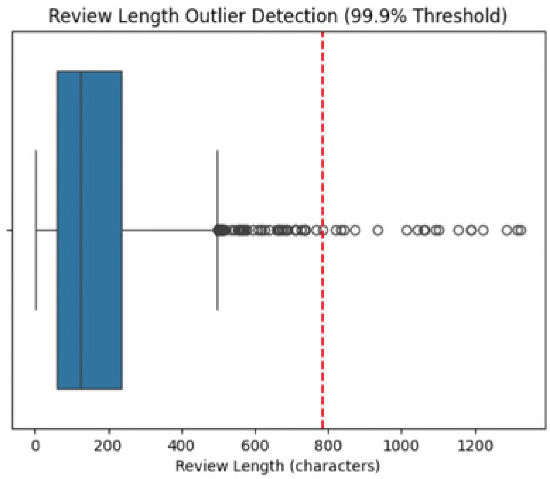
Findings

About 26% of reviews received a response from developers, while 74% did not receive a reply.

Developers tend to respond more often to lower ratings (1–2 stars), possibly to address user complaints or problems.

Review Length Outlier Detection

Finally, we analyzed the length of reviews to identify extreme values.



Findings

Most reviews are relatively short, but a small number are much longer and appear as outliers.

These longer reviews may include more detailed explanations or complaints from users.

Limitations of Secondary Data

Since the secondary dataset is pre exsiting, we cannot fully control how the data was originally collected or processed. This may introduce some inconsistencies or missing information in the dataset.

In addition, the dataset mainly represents English-speaking users and different cultural contexts. As a result, user behavior and feedback patterns may differ from those in the Arabic dataset used in the primary analysis.

Therefore, the results from the secondary dataset should be interpreted carefully, especially when comparing them with the primary data.

9.Comparison Between Primary and Secondary Data

In this section, we compared the results from the primary dataset (Arabic reviews collected for this study) and the secondary dataset (MHARD) which contains English reviews from a larger global dataset.

The comparison helps us understand the differences between the two datasets in terms of ratings, sentiment distribution, review length, and developer responses.

Key Metrics Comparison

To better understand the differences between the two datasets, several key metrics were compared including the total number of reviews, average rating, and the percentage of positive, negative, and neutral reviews.

	Metric	Primary (Our Study)	Secondary (MHARD Dataset)
0	Total Reviews	1031	18000
1	Language	Arabic	English
2	Average Rating	3.15	3.94
3	Positive Reviews (4-5 stars) %	50.8%	72.8%
4	Negative Reviews (1-2 stars) %	43.1%	21.7%
5	Neutral Reviews (3 stars) %	6.1%	5.5%

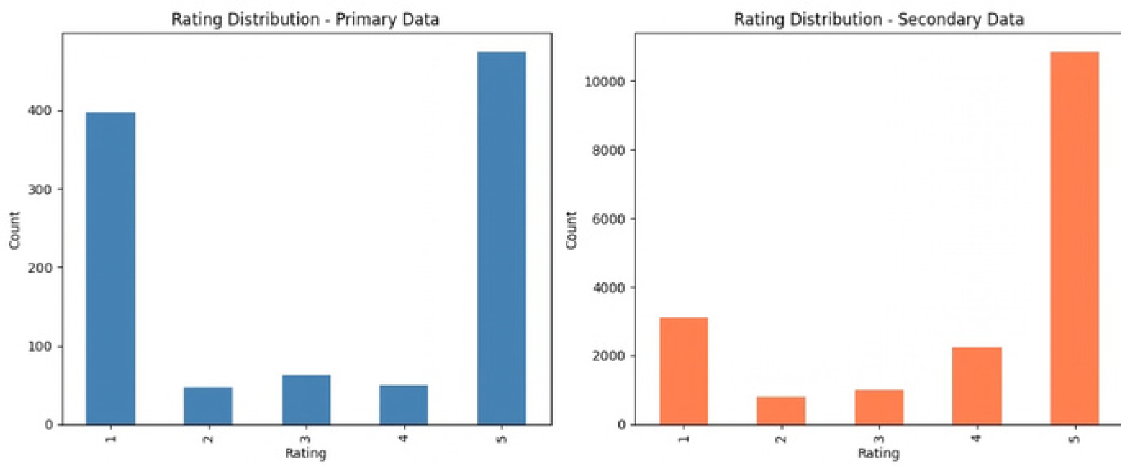
Findings

The primary dataset contains 1,031 Arabic reviews, while the secondary dataset contains 18,000 English reviews, making it much larger.

The average rating in the primary dataset is 3.15, while the secondary dataset has a higher average rating of 3.94.

In terms of sentiment distribution, the primary dataset has a higher percentage of negative reviews, while the secondary dataset contains more positive reviews.
This difference may indicate that users in the primary dataset report lower satisfaction with mental health apps compared to the global dataset.

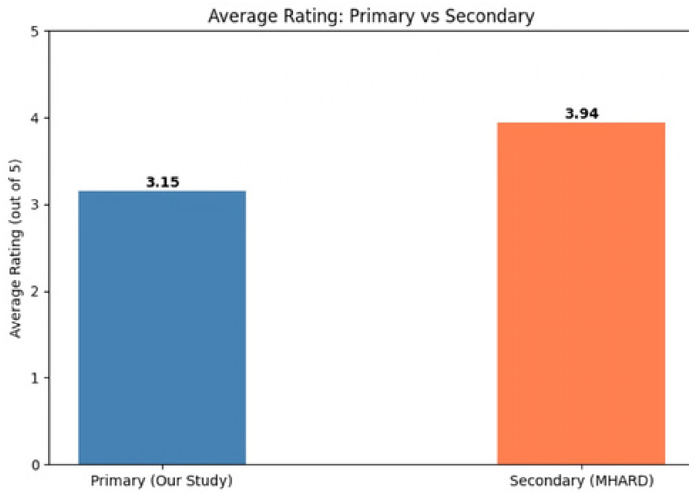
Rating Distribution



Findings

Both datasets show many 1-star and 5-star reviews, with fewer reviews in the middle ratings.
However, the primary dataset has more 1-star reviews, while the secondary dataset has more 5-star reviews.
This pattern is consistent with the literature and confirms that users tend to write reviews mainly during very positive or very negative experiences.

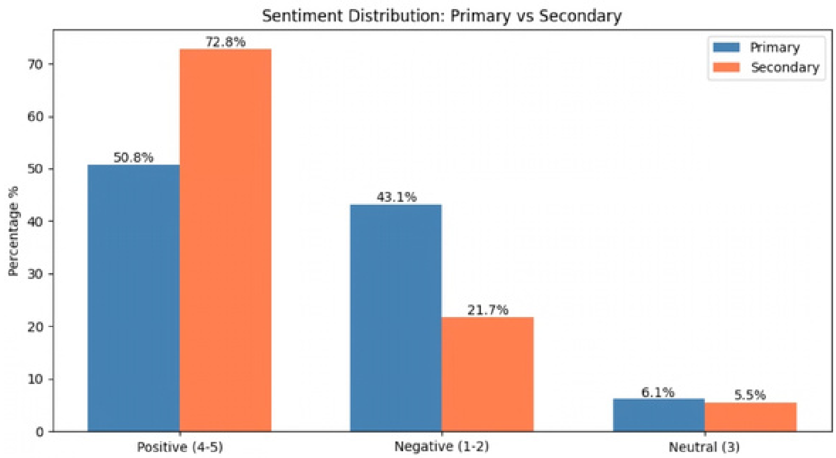
Average Rating



Findings

The secondary dataset has a higher average rating, suggesting more positive feedback compared to the primary dataset.This difference suggests that Arabic-speaking users rate mental health apps lower, possibly due to pricing issues, language barriers, and developing local apps.

Sentiment Distribution



Findings

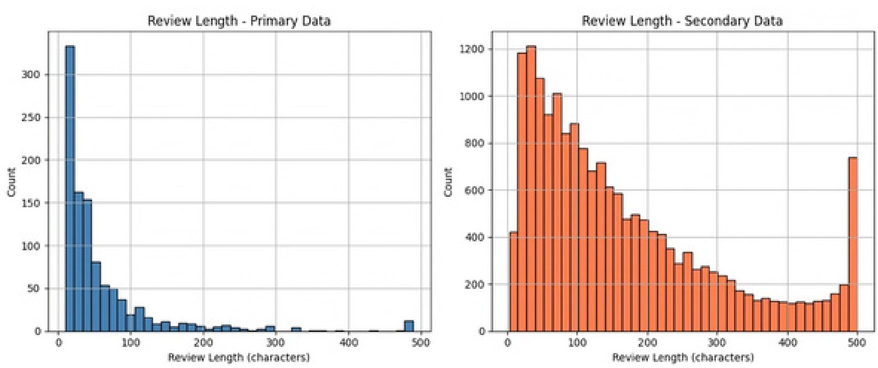
The primary dataset contains more negative reviews, while the secondary dataset has more positive reviews.

The neutral reviews are similar in both datasets.

This difference may be due to dataset differences, as the primary dataset contains Arabic reviews while the secondary dataset includes English reviews from a larger global sample.

King Saud University

Review Length Comparison

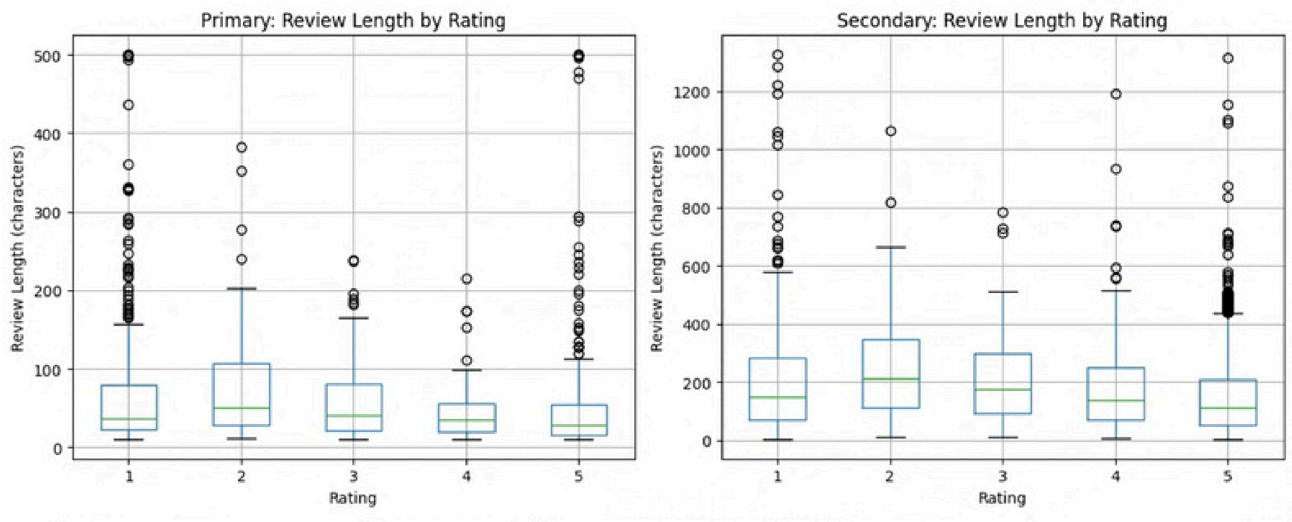



```
Primary avg review length  : 59.1 characters
Secondary avg review length: 167.7 characters
```

Findings

Both datasets show that most reviews are short.
However, the secondary dataset has longer reviews on average, while the primary dataset reviews are shorter and more direct.

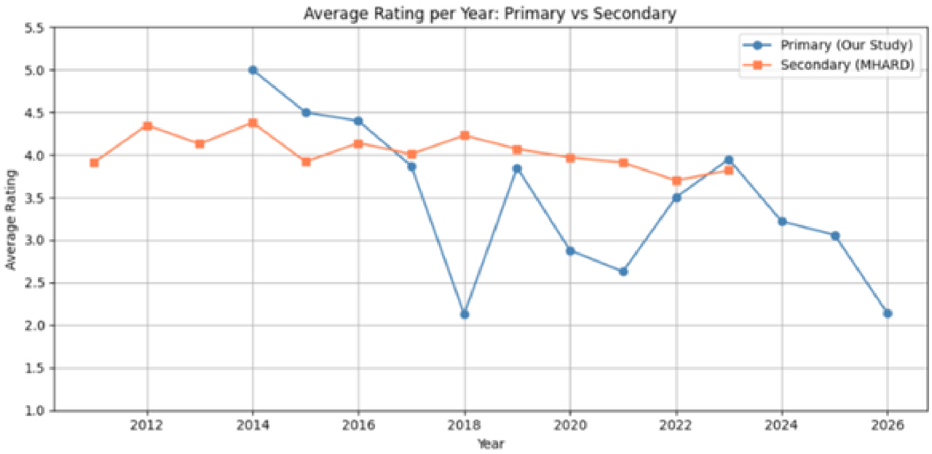
Review Length vs Rating



Findings

In both datasets, lower ratings tend to have longer reviews, meaning users usually write more when they are dissatisfied.

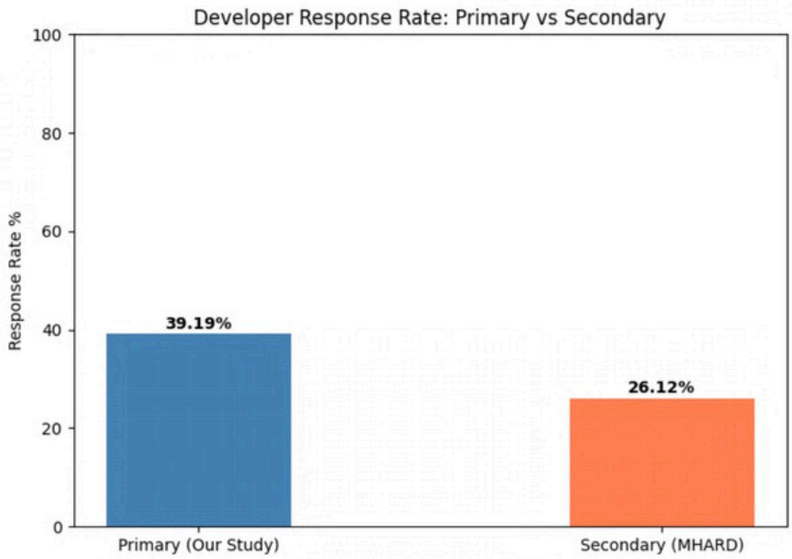
Average Rating Over Time



Findings

The secondary data shows improving ratings over time, while the primary data fluctuates with no clear trend. One possible explanation for this difference is the variation in dataset size.

Developer Response Rate



Findings

Developers respond to only some reviews in both datasets.

The primary dataset has a higher response rate (about 39%) compared to the secondary dataset (about 26%).

The difference may be due to dataset size since the secondary dataset covers many more apps with varying engagement levels.

Contextualizition of Findings

The secondary dataset helps us better understand the patterns observed in the primary data. We noticed that some patterns are similar in both datasets. Both show a bimodal rating distribution, where most reviews are either very positive (5 stars) or very negative (1 star). In addition, both datasets show a weak negative correlation between rating and review length, indicating that users tend to write longer reviews when they are dissatisfied.

At the same time, some differences appear between the datasets. The primary dataset has a lower average rating (3.16) and a higher percentage of negative reviews (43%) compared to the secondary dataset (3.94 average rating and 21.7% negative reviews). The secondary dataset also shows more stable ratings over time, while the primary dataset fluctuates across years. These differences may be related to variations in language, dataset size, and the applications included in each dataset.

10.Summary of New Insights and Hypotheses

From this **exploratory analysis**, several important insights were identified.

First, Arabic-speaking users appear to be slightly less satisfied compared to users in the global dataset. This may be related to differences in user expectations, app features, or language support.

Second, the rating distribution shows a polarized pattern, where many reviews are either very positive (5 stars) or very negative (1 star).

Third, negative reviews tend to be longer and more detailed, which suggests that dissatisfied users are more likely to explain their experiences.

Fourth, developer response rates differ between the two datasets, with the primary dataset showing a higher response rate.

Based on these observations, the following hypotheses were generated:

H1: Arabic-speaking users show lower overall satisfaction compared to users in the global dataset.
H2: User reviews tend to be polarized, with most ratings being either very positive or very negative.
H3: Lower-rated reviews tend to be longer than higher-rated reviews.
H4: Developer response rates differ between datasets and may reflect differences in developer engagement.

These hypotheses will be tested in Phase 3 using sentiment analysis and topic modeling.

11. Data Preprocessing

In this phase, several preprocessing steps were applied to prepare the dataset for modelling.

Step 1: Labeling

First, sentiment labels were created based on rating values. Ratings of 1 and 2 were mapped to negative, rating 3 to neutral, and ratings of 4 and 5 to positive.

Step 2: Text Cleaning and Normalization

Next, the Arabic text was cleaned and normalized to improve data quality. This included removing non-Arabic characters, punctuation, numbers, and unnecessary symbols. Arabic letters were normalized (e.g., different forms of letters were unified) to reduce variation in the text. and extra spaces were handled to ensure consistency. Rows with missing, empty, or zero-length text were removed to ensure valid input for the models.

Step 3: Data Preparation and Splitting

Finally, the dataset was prepared in a consistent format and split into training and testing sets using an 80/20 ratio. Stratified sampling was applied to preserve the distribution of sentiment classes in both sets and ensure fair model evaluation.

12. Modelling

This project is formulated as a multi-class classification task to classify Arabic user reviews into three sentiment categories: positive, neutral, and negative.

All models were trained and evaluated using the same dataset, preprocessing steps, and train-test split to ensure a fair comparison between models.

TF-IDF vectorization was used to convert text data into numerical features. Only the training data was used to fit the vectorizer to prevent data leakage.

12.1 Baseline Model: Logistic Regression

Logistic Regression was used as a baseline model due to its effectiveness and simplicity in text classification tasks. It works well with high-dimensional sparse data such as TF-IDF features.

The model achieved an accuracy of 78.2% and a macro F1-score of 0.54. The results show that the model performs well in classifying positive and negative reviews - (fails to correctly classify neutral reviews.)

12.2 Model 2: Random Forest

Random Forest is an ensemble learning method that builds multiple decision trees and combines their predictions using majority voting.

The model achieved an accuracy of 71.0% and a macro F1-score of 0.48. The performance was lower than Logistic Regression, indicating that tree-based models may not perform well with sparse text features like TF-IDF. _ Additionally, the model fails to correctly classify neutral reviews.

12.3 Model 3: XGBoost

XGBoost is a gradient boosting algorithm that builds models sequentially to improve performance. It is known for its strong performance in many machine learning tasks.

For XGBoost, sentiment labels were encoded into numerical values (negative=0, neutral=1, positive=2).

The model achieved an accuracy of 70.5% and a macro F1-score of 0.52. Although it performed better than Random Forest, it still did not outperform Logistic Regression. Unlike the other models, XGBoost showed some ability to detect neutral reviews, though performance remains low.

13. Model Evaluation and Discussion

Among all evaluated models, Logistic Regression achieved the best overall performance, with an accuracy of 78.26% and a macro F1-score of 0.5385.

The results indicate that the model performs well in classifying positive and negative sentiments. However, it fails to classify neutral reviews, which are often misclassified.

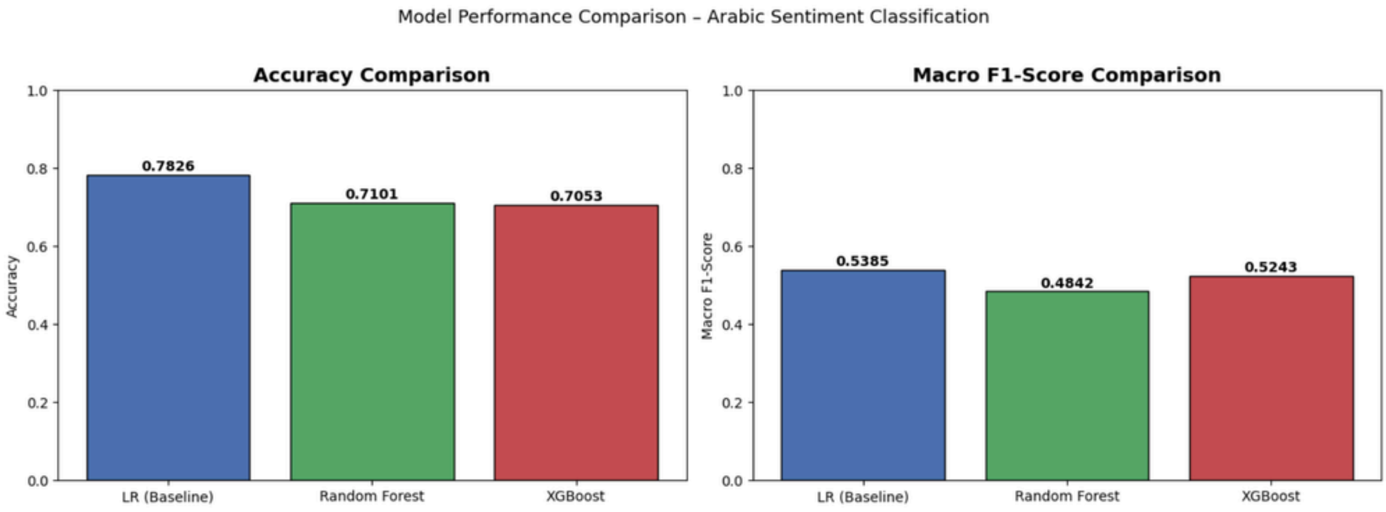
This behavior is expected due to two main reasons. First, the number of neutral samples is relatively small compared to positive and negative classes, leading to class imbalance. Second, neutral reviews tend to have less distinctive linguistic patterns, making them harder to separate clearly.

The confusion matrix confirms that neutral reviews are the most difficult class to predict accurately.

This finding is consistent with the earlier Exploratory Data Analysis (EDA), where extreme ratings (1 and 5) were more dominant. These extreme classes are easier for the model to learn and classify.

When comparing all models, Logistic Regression outperformed both Random Forest and XGBoost in terms of both accuracy and macro F1-score.

Therefore, Macro F1-score was used as the primary metric for model selection due to class imbalance.



The figure shows a comparison of model performance in terms of Accuracy and Macro F1-score.

Logistic Regression achieved the highest performance among all models, with the best accuracy and Macro F1-score.

Random Forest showed lower performance, while XGBoost performed better than Random Forest in terms of Macro F1-score but slightly lower in accuracy compared to Logistic Regression.

This visualization clearly shows that Logistic Regression provides the best overall performance, mainly due to its strong classification of positive and negative reviews, while performance on neutral reviews remains weak across all models.

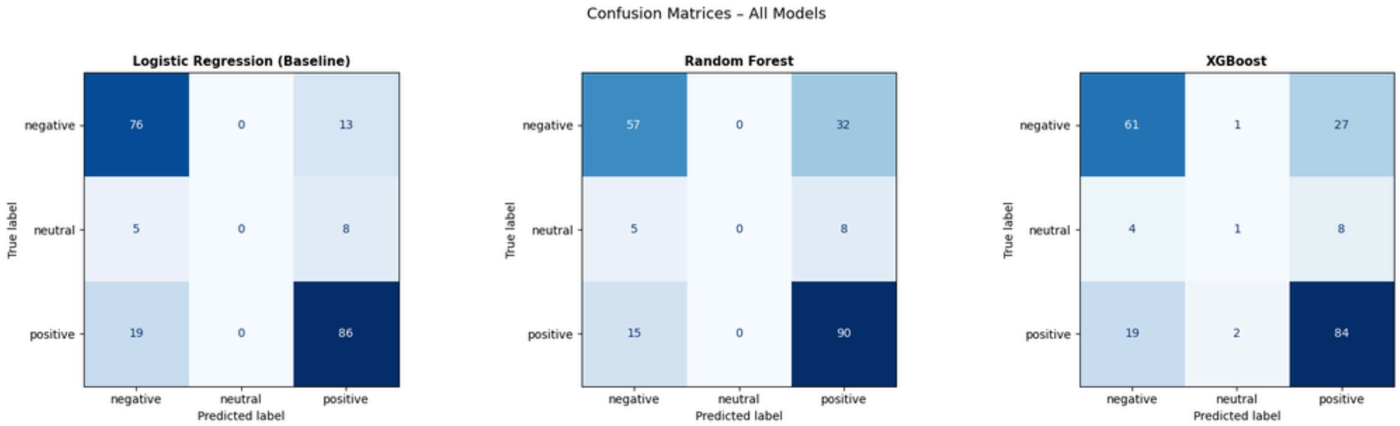
Overall, the results provide several important findings.

First, the analysis confirms that extreme sentiments (positive and negative) are easier to classify compared to neutral reviews.

Second, the class imbalance in the dataset affects model performance, especially for the neutral class.

Third, simpler models such as Logistic Regression can outperform more complex models when working with TF-IDF features.

Finally, the findings highlight the importance of improving feature representation and balancing the dataset in future work to enhance model performance.


Confusion Matrices – All Models
Logistic Regression (Baseline)
Random Forest
XGBoost
True label
Predicted label
negative
neutral
positive
76
0
13
5
0
8
19
0
86
57
0
32
5
0
8
15
0
90
61
1
27
4
1
8
19
2
84

14. Conclusion

In this phase, multiple machine learning models were developed to classify Arabic user reviews into sentiment categories (positive, neutral, and negative).

Among all models, Logistic Regression achieved the best overall performance, making it the most suitable model for this task. This result highlights the effectiveness of simple linear models when working with high-dimensional text features such as TF-IDF.

However, the results also reveal a key challenge: neutral reviews are more difficult to classify accurately. This is mainly due to class imbalance and the less distinctive linguistic patterns of neutral sentiment.

Overall, this study demonstrates the importance of proper text preprocessing, feature representation, and model selection in sentiment analysis tasks. Future work could focus on improving performance by using more advanced text representations, handling class imbalance, and exploring deep learning approaches.

13.1 Hypothesis Evaluation

The four hypotheses proposed during Phase 2 exploratory analysis were evaluated based on the modelling results and EDA findings:

H1 — Confirmed: Arabic-speaking users showed lower overall satisfaction with mental health applications. The primary dataset recorded an average rating of 3.15 compared to 3.94 in the secondary dataset, with 43.1% negative reviews versus 21.7% in the global dataset. This suggests that Arabic-speaking users have higher unmet expectations from mental health apps.

H2 — Confirmed: User reviews tend to be polarized. The classification results support this hypothesis — all three models consistently achieved their lowest F1-scores on the neutral class, while positive and negative classes were classified more accurately. This reflects the bimodal rating distribution observed in the EDA, where most reviews are either 1-star or 5-star.

H3 — Confirmed: Lower-rated reviews tend to be longer. The negative correlation between rating and review length was observed in both datasets (Primary: -0.127, Secondary: -0.149), indicating that dissatisfied users write more detailed feedback to explain their negative experiences.

H4 — Confirmed: Developer response rates differ betweendatasets. The primary dataset showed a response rate of 39.19% compared to 26.12% in the secondary dataset, suggesting that developers of Arabic-targeted apps may be more actively engaged with user feedback.